

Tutorial: Phase 1 - Project Setup & Database Design

Introduction

This tutorial provides step-by-step instructions for **setting up the MySQL database** and **local development environment** for the **Pandemic Resilience System (PRS)**.

The Database schema is indicative and cannot be applied directly to PRS

Objectives:

- Define project requirements & scope
- Set up MySQL database with required tables
- Create an **ER diagram** & database schema
- Develop test data scripts for initial database population
- Set up a local **XAMPP/LAMP development environment**

Step 1: Define Project Requirements & Scope

Pandemic Resilience System (PRS) is designed to manage:

- **Users:** Government officials, merchants, and public members
- **Vaccination Records:** Track immunization details
- **Supply Chain Management:** Merchant-related data
- **Security Measures:** Encryption keys, audit logs
- **Unstructured Data Support:** PDFs, images

These requirements guide the **database structure** and **system architecture**.

Step 2: Set Up MySQL Database

Create a New Database

1. Open **phpMyAdmin** (if using XAMPP) or connect via MySQL CLI:

```
CREATE DATABASE prs_database;
```

2. Select the database:

```
USE prs_database;
```

Create Required Tables

1 Users Table

```
CREATE TABLE users (  
    user_id INT PRIMARY KEY AUTO_INCREMENT,  
    full_name VARCHAR(255) NOT NULL,  
    email VARCHAR(255) UNIQUE NOT NULL,  
    password_hash VARCHAR(255) NOT NULL,  
    phone VARCHAR(20),  
    national_id VARCHAR(50) UNIQUE NOT NULL,  
    prs_id VARCHAR(50) UNIQUE NOT NULL,  
    role_id INT NOT NULL,  
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
    FOREIGN KEY (role_id) REFERENCES roles(role_id) ON DELETE CASCADE  
);
```

2 Roles Table

```
CREATE TABLE roles (  
    role_id INT PRIMARY KEY AUTO_INCREMENT,  
    role_name VARCHAR(50) NOT NULL UNIQUE,  
    description TEXT  
);
```

3 Vaccination Records Table

```
CREATE TABLE vaccination_records (  
    record_id INT PRIMARY KEY AUTO_INCREMENT,  
    user_id INT NOT NULL,  
    vaccine_name VARCHAR(255) NOT NULL,  
    date_administered DATE NOT NULL,  
    dose_number INT NOT NULL,  
    provider VARCHAR(255),  
    lot_number VARCHAR(50),  
    expiration_date DATE,  
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
    FOREIGN KEY (user_id) REFERENCES users(user_id) ON DELETE CASCADE  
);
```

4 Documents Table (For PDFs & Images)

```
CREATE TABLE documents (  
    document_id INT PRIMARY KEY AUTO_INCREMENT,  
    file_type VARCHAR(50) NOT NULL,  
    file_path TEXT NOT NULL,  
    uploaded_by INT NOT NULL,
```

```

        related_record_id INT,
        upload_date TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
        FOREIGN KEY (uploaded_by) REFERENCES users(user_id) ON DELETE
CASCADE,
        FOREIGN KEY (related_record_id) REFERENCES
vaccination_records(record_id) ON DELETE SET NULL
    );

```

5 Audit Logs Table

```

CREATE TABLE audit_logs (
    log_id INT PRIMARY KEY AUTO_INCREMENT,
    user_id INT NOT NULL,
    action VARCHAR(255) NOT NULL,
    entity_affected VARCHAR(100),
    record_id INT,
    timestamp TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (user_id) REFERENCES users(user_id) ON DELETE CASCADE
);

```

6 Encryption Keys Table

```

CREATE TABLE encryption_keys (
    key_id INT PRIMARY KEY AUTO_INCREMENT,
    key_type VARCHAR(50) NOT NULL,
    key_value TEXT NOT NULL,
    owner VARCHAR(50),
    created_date TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

```

Step 3: Create an ER Diagram

1. Use **MySQL Workbench** or **draw.io** to create an ER diagram.
2. Include tables **Users, Roles, Vaccination Records, Documents, Audit Logs, and Encryption Keys**.
3. Define **relationships (Foreign Keys)** between tables.

Step 4: Develop Test Data Scripts

Insert Sample Roles

```

INSERT INTO roles (role_name, description) VALUES

```

```
('Government Official', 'Can monitor vaccination and supply chain'),
('Merchant', 'Can manage pandemic-related supplies'),
('Public Member', 'Can track vaccination records');
```

Insert Sample Users

```
INSERT INTO users (full_name, email, password_hash, phone, national_id,
prs_id, role_id) VALUES
('Alice Johnson', 'alice@gov.org', SHA2('securepassword', 256),
'+1234567890', 'GOV12345', 'PRS001', 1),
('Bob Merchant', 'bob@merchant.com', SHA2('merchantpass', 256),
'+1987654321', 'MER12345', 'PRS002', 2);
```

Insert Sample Vaccination Records

```
INSERT INTO vaccination_records (user_id, vaccine_name,
date_administered, dose_number, provider, lot_number, expiration_date)
VALUES
(1, 'COVID-19 Vaccine AstraZeneca', '2023-01-10', 1, 'City Health
Clinic', 'AZ-12345', '2023-12-31');
```

Step 5: Set Up Local Development Environment

Install XAMPP (Windows/macOS/Linux)

1. Download **XAMPP** from [Apache Friends](https://www.apachefriends.org/).
2. Install and open the **XAMPP Control Panel**.
3. Start **Apache & MySQL**.
4. Open **phpMyAdmin** (<http://localhost/phpmyadmin>) and create prs_database.

Deliverables

- ✓ Database schema with SQL scripts
- ✓ ER diagram & data models
- ✓ Local MySQL instance with sample data